

Distributed Algorithm for Principal Component Analysis

Thierry Sae-Lim

Under the supervision of Professor Norikazu TAKAHASHI

Contents

Context of the internship

Introduction

Multi-Agent Systems

Principal Component Analysis

Decentralized Algorithm for Principal Component Analysis

Numerical Experiments

Conclusion and Perspectives

Context of the internship

M1 - Artificial Intelligence, Algorithm, Interaction, Decision Making (AI2D)

Science, medicine, and humanities

Internship undertaken within the framework of partnership between the universities



Introduction

Increasing amount of data generated and processed + sensitive data

→ distribution of data across many devices and use of distributed algorithms.

PCA : dimensionality reduction technique requires centralized information

How to compute principal components without sharing data ?



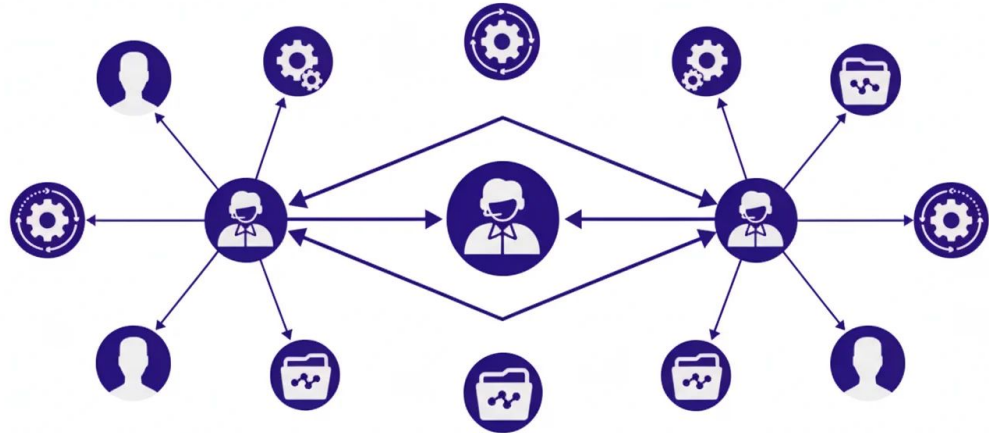
Multi-Agent Systems

Introduction

Centralized Multi-Agent System



Multi Agent System



Multi-Agent Systems - Distributed Averaging

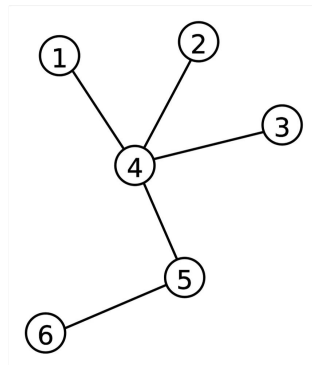
Distributed Linear Iteration:

$$x_i(t+1) = W_{ii}x_i(t) + \sum_{j \in i_n} W_{ij}x_j(t), \quad i = 1, \dots, n$$

Weight Matrix W :

$$W_{ij} = \frac{1}{1 + \max\{i_d, j_d\}}, \quad \{i, j\} \in E$$

$$W_{ii} = 1 - \sum_{j \in V, j \neq i} W_{ij}, \quad i = 1, \dots, N_{agent}$$



$$W = \begin{pmatrix} 0.8 & 0 & 0 & 0.2 & 0 & 0 \\ 0 & 0.8 & 0 & 0.2 & 0 & 0 \\ 0 & 0 & 0.8 & 0.2 & 0 & 0 \\ 0.2 & 0.2 & 0.2 & 0.2 & 0.2 & 0 \\ 0 & 0 & 0 & 0.2 & 0.47 & 0.33 \\ 0 & 0 & 0 & 0 & 0.33 & 0.67 \end{pmatrix}$$

Algorithm 2 Distributed Averaging

Require: $G, W, x_1(0), \dots, x_{N_{agent}}(0), \varepsilon_{DA}, T_{DA}$

Ensure: Estimates of the mean $\bar{x}(0)$ for all agents

1: $t \leftarrow 0$

2: **while not** $((\forall i \in V : |x_i(t) - \bar{x}(0)| < \varepsilon_{DA}) \text{ or } (t \geq T_{DA}))$ **do**

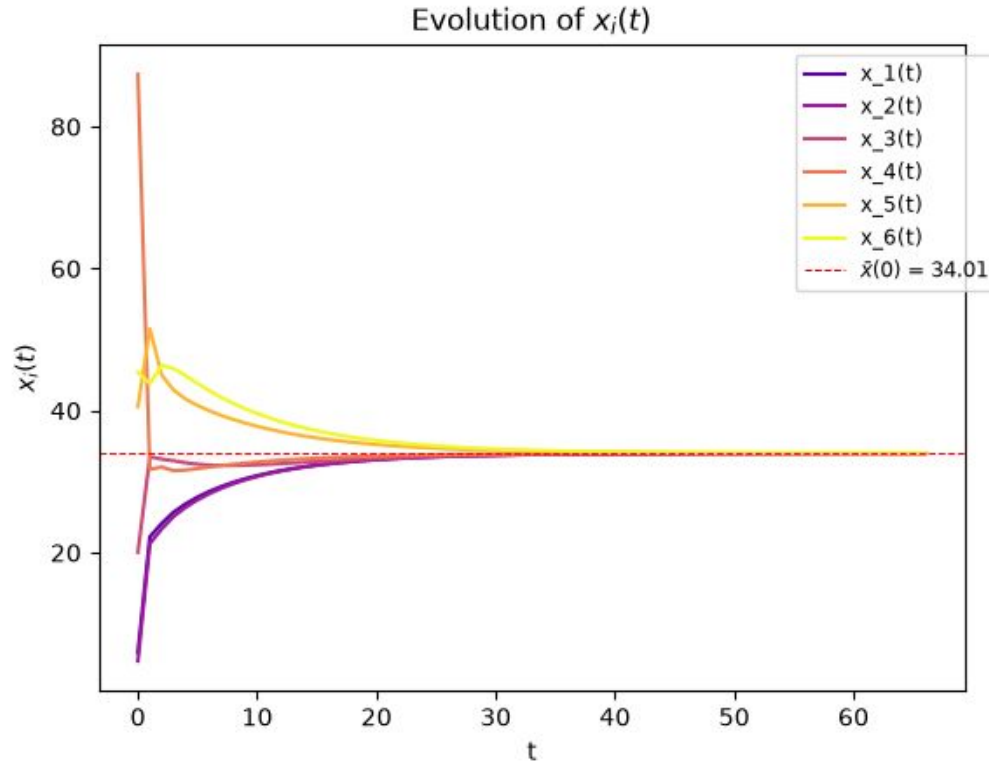
3: $x(t+1) \leftarrow Wx(t)$

4: $t \leftarrow t + 1$

5: **end while**

6: **return** $x_1(t), \dots, x_{N_{agent}}(t)$

Multi-Agent Systems - Distributed Averaging - Test

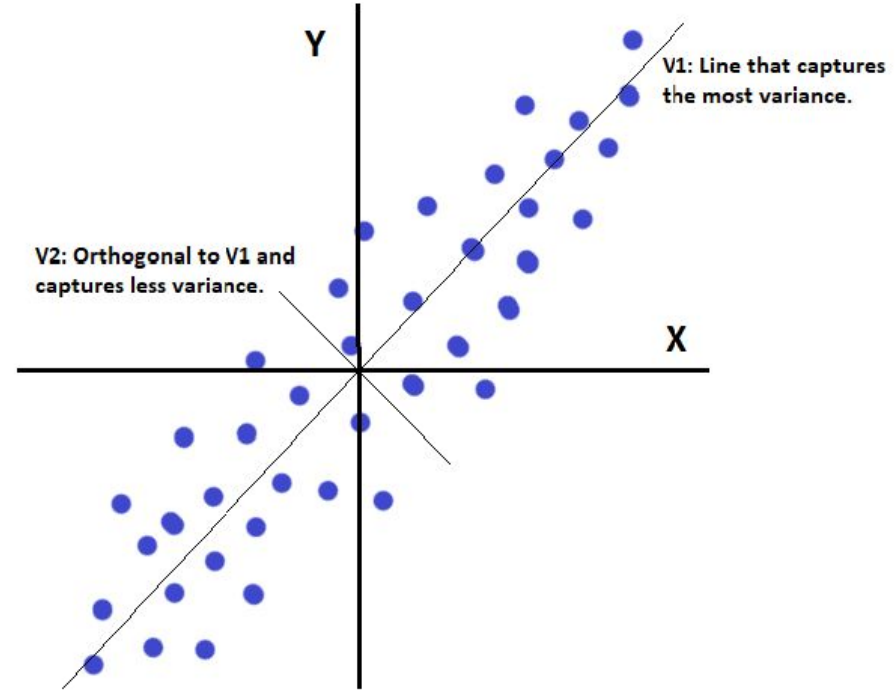


Principal Component Analysis - Introduction

Statistical technique used for analyzing and reducing the dimensionality of dataset $D = (d_1, d_2, \dots, d_{L_{dim}}) \in \mathbb{R}^{N_{dim} \times L_{dim}}$.

Finds a subspace of dimension P_{dim} that minimize the information loss.

→ Compute the eigenvectors of the dataset covariance matrix.



Principal Component Analysis - Power Method

Algorithm 5 Power Method

Require: $D, u_1(0), \dots, u_{P_{dim}}(0), T_{PC}$

Ensure: $U(T_{PC})$

```
1:  $S_1 \leftarrow \frac{1}{L_{dim}} D D^\top$ 
2: for  $p = 1, \dots, P_{dim}$  do
3:   for  $t = 0, \dots, T_{PC} - 1$  do
4:      $u_p(t+1) \leftarrow \frac{S_p u_p(t)}{\|S_p u_p(t)\|}$ 
5:   end for
6:    $S_{p+1} \leftarrow S_p - S_p u_p(T_{PC}) u_p(T_{PC})^\top$ 
7: end for
8: return  $u_1(T_{PC}), \dots, u_{P_{dim}}(T_{PC})$ 
```

- Compute the principal components sequentially
- Propagation of approximation errors

Principal Component Analysis - Update Rule

Algorithm 6 Update Rule

Require: $D, u_1(0), \dots, u_{P_{dim}}(0), T_{PC}, k_1, k_2, \epsilon$

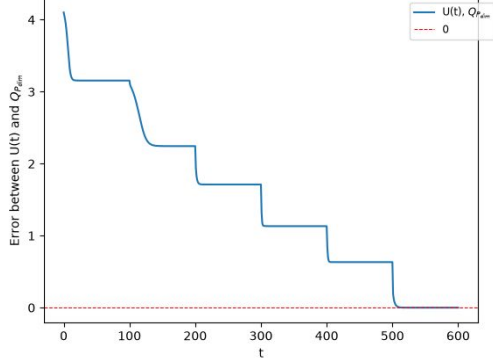
Ensure: $U(T_{PC})$

```
1:  $S \leftarrow \frac{1}{L_{dim}} DD^\top$ 
2: for  $t = 0, \dots, T_{PC} - 1$  do
3:   for  $p = 1, \dots, P_{dim}$  do
4:      $u_p(t+1) \leftarrow u_p(t) + \epsilon \left( -k_1 \left( \sum_{j=1}^P u_j(t) u_j(t)^\top u_p(t) \right) + k_2 S u_p(t) \right)$ 
5:   end for
6:   for  $p = 1, \dots, P_{dim}$  do
7:      $u_p(t+1) \leftarrow \frac{u_p(t+1)}{\|u_p(t+1)\|}$ 
8:   end for
9: end for
10: return  $u_1(T_{PC}), \dots, u_{P_{dim}}(T_{PC})$ 
```

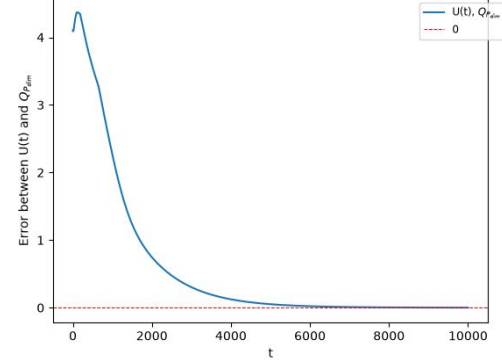
- Compute the principal components in parallel

Principal Component Analysis - Tests

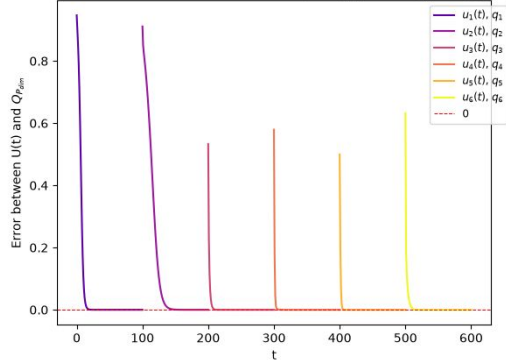
Evolution of the error between $U(t) = (u_1 \dots u_{p_{dim}})$ and $Q_{p_{dim}} = (q_1 \dots q_{p_{dim}})$



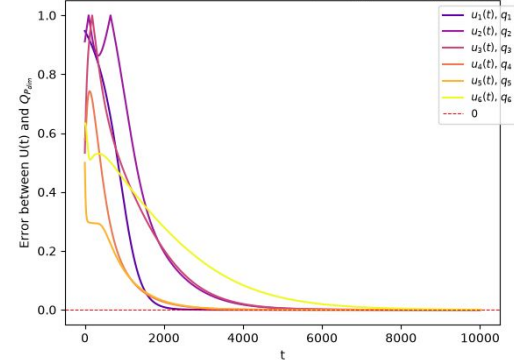
Evolution of the error between $U(t) = (u_1 \dots u_{p_{dim}})$ and $Q_{p_{dim}} = (q_1 \dots q_{p_{dim}})$



Evolution of the error between $U(t) = (u_1 \dots u_{p_{dim}})$ and $Q_{p_{dim}} = (q_1 \dots q_{p_{dim}})$



Evolution of the error between $U(t) = (u_1 \dots u_{p_{dim}})$ and $Q_{p_{dim}} = (q_1 \dots q_{p_{dim}})$



Decentralized Algorithm for PCA - Introduction

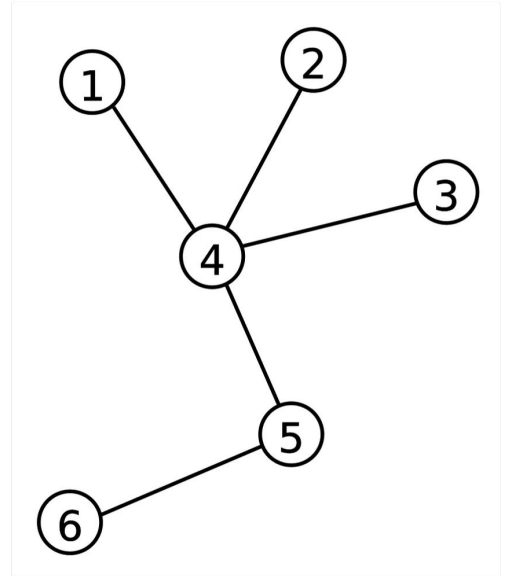
Network structure G

Weight Matrix W

Global data matrix $D = (D_1, D_2, \dots, D_{N_{\text{agent}}})$

For each agent:

- Data matrices $D_m \in \mathbb{R}^{N_{\text{dim}} \times L_m}$
- Initial vectors $U_m(0) = (u_{1,m}(0), u_{2,m}(0), \dots, u_{P_{\text{dim}},m}(0)) \in \mathbb{R}^{N_{\text{dim}} \times P_{\text{dim}}}$.



Decentralized Algorithm for PCA - Algorithm

Step 1:

Consensus on vectors $u_{p,m}(t)$ to obtain vector $y_{p,m}(t)$.

Step 2:

Consensus on $D_m D_m^\top y_{p,m}(t)$ to obtain vector $z_{p,m}(t)$.

Step 3:

Performing an iteration of the Update Rule for each agent.

Algorithm 10 Decentralized PCA (DPCA)

Require: $G, W, D_1, \dots, D_{N_{agent}}, U_1(0), \dots, U_{N_{agent}}(0), \varepsilon_{DA}, T_{DAy}, T_{DAz}, \varepsilon_{PC}, T_{PC}, k_1, k_2, \epsilon$

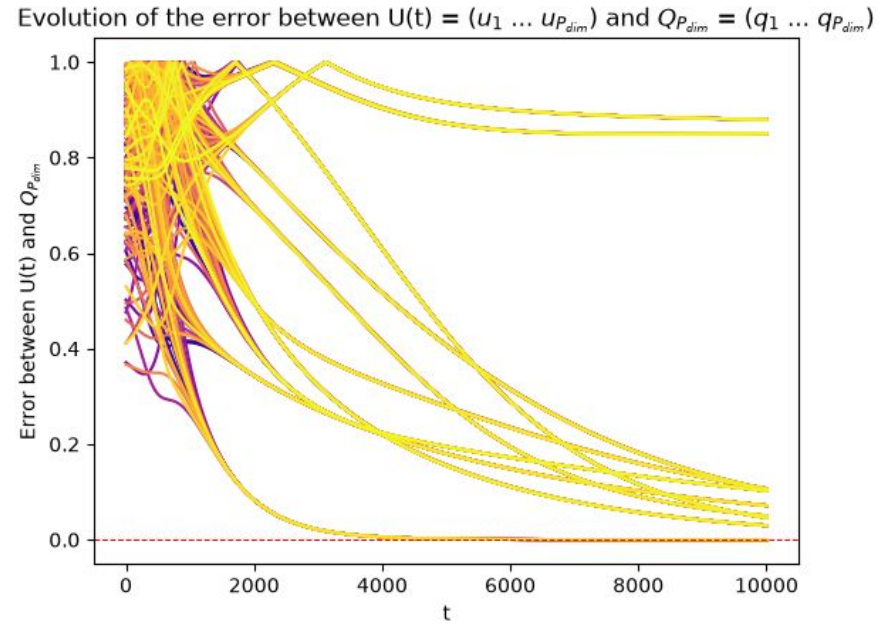
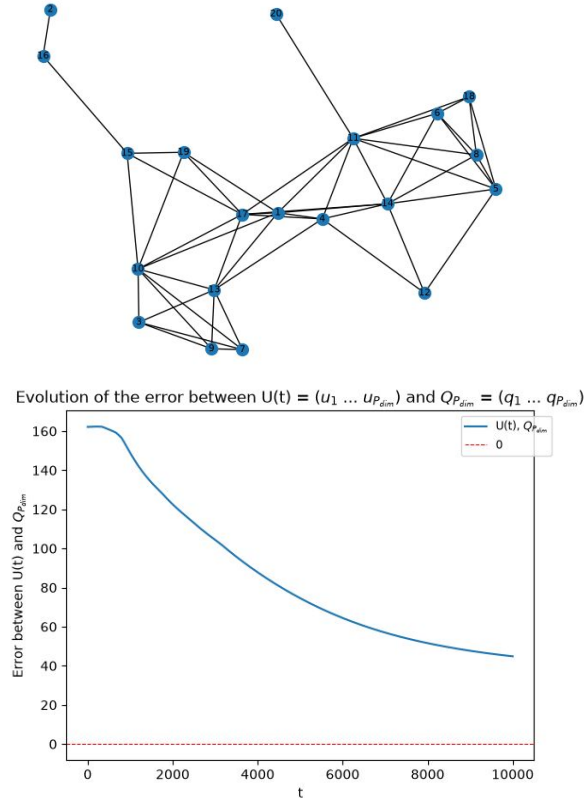
Ensure: Approximation of the principal components of D for all agents

```

1:  $t \leftarrow 0$ 
2: while not  $((\forall m \in V : \mathcal{E}_m(t) < \varepsilon_{PC}) \text{ or } (t \geq T_{PC}))$  do ▷ Step 1
3:   for  $p = 1, \dots, P_{dim}$  do
4:      $(x_1, \dots, x_{N_{agent}}) \leftarrow u_{p,1}(t), \dots, u_{p,N_{agent}}(t)$ 
5:      $(y_{p,1}(t), \dots, y_{p,N_{agent}}(t)) \leftarrow$ 
       DISTRIBUTEDAVERAGING( $G, W, x_1, \dots, x_{N_{agent}}, \varepsilon_{DA}, T_{DAy}$ )
6:   end for ▷ Step 2
7:   for  $p = 1, \dots, P_{dim}$  do
8:      $(x_1, \dots, x_{N_{agent}}) \leftarrow D_1 D_1^\top y_{p,1}(t), \dots, D_{N_{agent}} D_{N_{agent}}^\top y_{p,N_{agent}}(t)$ 
9:      $(z_{p,1}(t), \dots, z_{p,N_{agent}}(t)) \leftarrow$ 
       DISTRIBUTEDAVERAGING( $G, W, x_1, \dots, x_{N_{agent}}, \varepsilon_{DA}, T_{DAz}$ )
10:  end for ▷ Step 3
11:  for  $m = 1, \dots, N_{agent}$  do
12:    for  $p = 1, \dots, P_{dim}$  do
13:       $u_{p,m}(t+1) \leftarrow$ 
14:         $u_{p,m}(t) + \epsilon \left( -k_1 \left( \sum_{j=1}^p y_{j,m}(t) y_{j,m}(t)^\top y_{p,m}(t) \right) + k_2 \frac{N_{agent}}{L_{dim}} z_{p,m}(t) \right)$ 
15:    end for
16:    for  $p = 1, \dots, P_{dim}$  do
17:       $u_{p,m}(t+1) \leftarrow \frac{u_{p,m}(t+1)}{\|u_{p,m}(t+1)\|}$ 
18:    end for
19:  end for
20:   $t \leftarrow t + 1$ 
21: end while
22: return  $U_1(t), \dots, U_{N_{agent}}(t)$ 

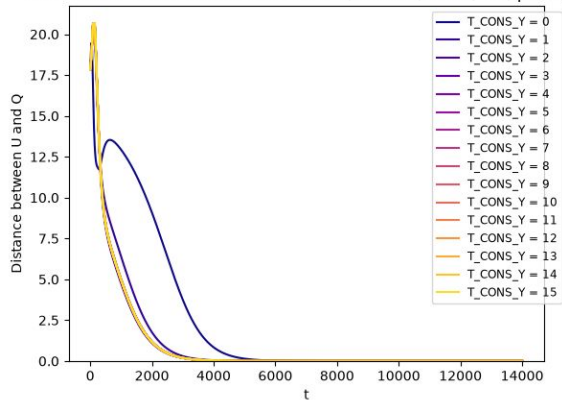
```

Decentralized Algorithm for PCA - Test

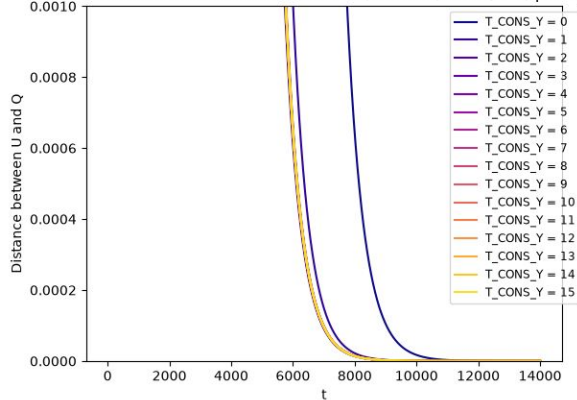


Numerical Experiments - Accuracy and Iterations

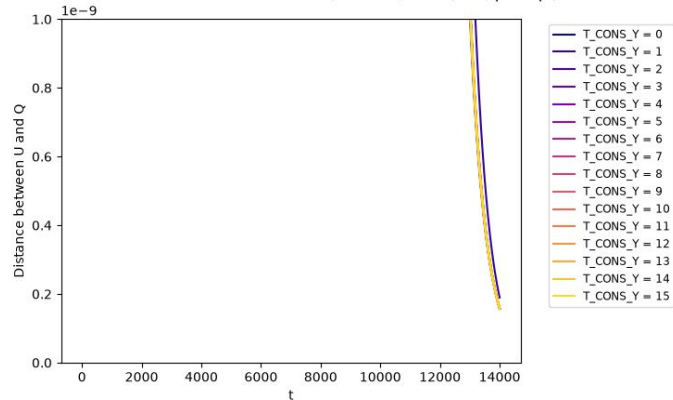
Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



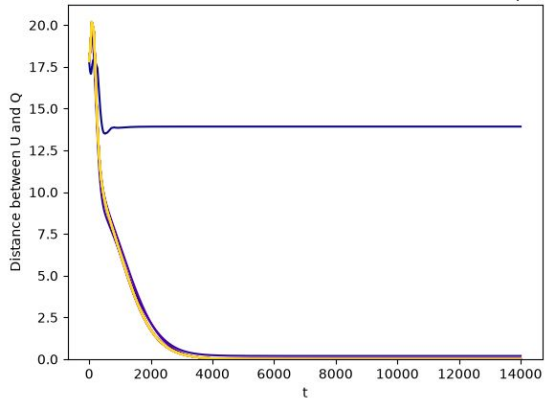
Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



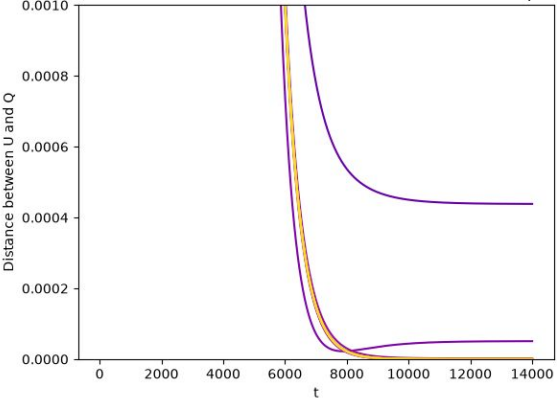
Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



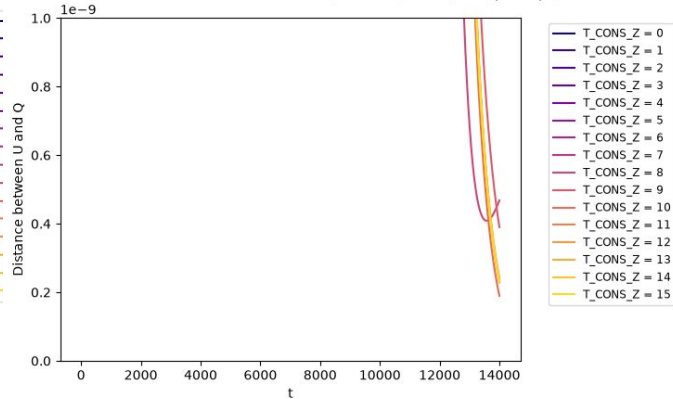
Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



Evolution of the distance between $U = (u_1 \dots u_P)$ and $Q' = (q_1 \dots q_P)$



Numerical Experiments - Optimal Iteration Parameters

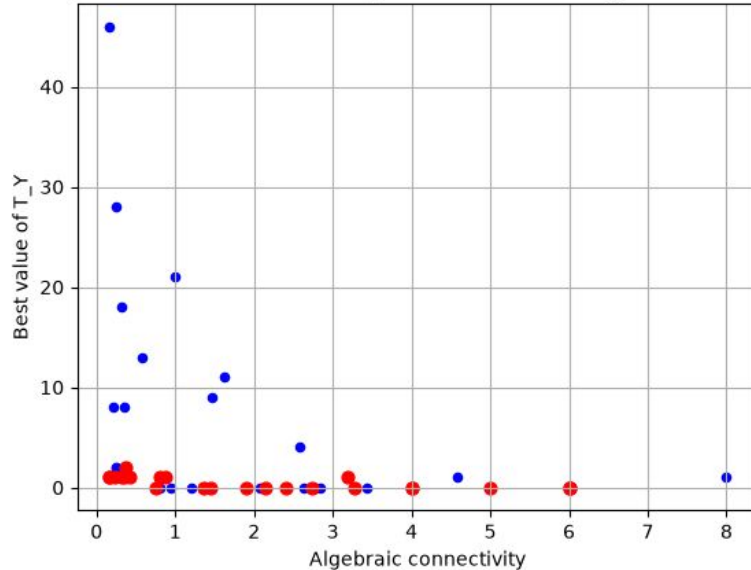
Network Structures

T_Z_T_Y	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
10	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	43082 33414 0 12	39116 27941 1 12	37755 25171 2 12	34576 21611 3 12	29512 17361 4 12	35370 19651 5 12	40128 21121 6 12	43380 21691 7 12	46074 21941 8 12	48532 22061 9 12	50853 22111 10 12	53136 22141 11 12	55375 22151 12 12	56705 22161 13 12	57555 22161 14 12	58299 22161 15 12	58947 22161 16 12	59502 22161 17 12	59961 22161 18 12	60321 22161 19 12	60592 22161 20 12
13	44044 31446 0 13	39180 26121 1 13	36896 23061 2 13	32708 19241 3 13	29340 16301 4 13	32832 17281 5 13	36800 18401 6 13	39606 18861 7 13	41932 19061 8 13	44068 19161 9 13	46080 19201 10 13	48050 19221 11 13	49336 19231 12 13	50301 19241 13 13	51136 19241 14 13	51875 19241 15 13	52521 19241 16 13	53074 19241 17 13	53533 19241 18 13	53892 19241 19 13	54164 19241 20 13
14	46515 31014 0 14	41408 25881 1 14	38726 22781 2 14	34074 18931 3 14	30609 16111 4 14	33660 16981 5 14	37821 18011 6 14	40590 18451 7 14	42895 18651 8 14	44976 18741 9 14	46975 18791 10 14	48345 18811 11 14	49400 18821 12 14	50326 18821 13 14	51155 18821 14 14	51918 18831 15 14	52563 18831 16 14	53115 18831 17 14	53574 18831 18 14	53933 18831 19 14	54205 18831 20 14
15	49344 30841 0 15	43945 25851 1 15	40932 22741 2 15	35891 18891 3 15	32100 16051 4 15	35490 16901 5 15	39380 17901 6 15	42159 18331 7 15	44472 18531 8 15	46550 18621 9 15	48037 18661 10 15	49238 18681 11 15	50279 18681 12 15	51224 18701 13 15	52052 18701 14 15	52788 18701 15 15	53432 18701 16 15	53984 18701 17 15	54442 18701 18 15	54802 18701 19 15	55073 18701 20 15
16	52292 30761 0 16	46512 25841 1 16	43168 22721 2 16	37740 18871 3 16	33663 16031 4 16	37114 16871 5 16	41078 17861 6 16	43896 18291 7 16	46200 18481 8 16	47879 18571 9 16	49289 18621 10 16	50458 18641 11 16	51493 18651 12 16	52411 18651 13 16	53237 18651 14 16	53972 18651 15 16	54641 18651 16 16	55193 18651 17 16	55652 18651 18 16	56011 18651 19 16	56283 18651 20 16
17	55314 30731 0 17	49077 25831 1 17	45440 22721 2 17	39606 18861 3 17	35244 16021 4 17	38778 16861 5 17	42816 17841 6 17	45875 18271 7 17	47672 18461 8 17	49281 18551 9 17	50642 18601 10 17	52037 18621 11 17	53257 18631 12 17	54382 18631 13 17	55362 18641 14 17	56288 18641 15 17	57156 18641 16 17	57983 18641 17 17	58697 18641 18 17	59356 18641 19 17	59972 18641 20 17
18	58349 30711 0 18	51660 25831 1 18	47691 22711 2 18	41470 18851 3 18	36823 16011 4 18	40464 16861 5 18	44600 17841 6 18	47204 18261 7 18	49167 18461 8 18	50766 18551 9 18	52090 18591 10 18	53250 18611 11 18	54284 18621 12 18	55200 18631 13 18	56052 18631 14 18	56787 18631 15 18	57430 18631 16 18	57983 18631 17 18	58441 18631 18 18	58800 18631 19 18	59072 18631 20 18
19	61420 30711 0 19	54243 25831 1 19	49962 22711 2 19	43355 18851 3 19	38424 16011 4 19	42150 16861 5 19	46132 17831 6 19	48750 18261 7 19	50686 18461 8 19	52269 18541 9 19	53592 18581 10 19	54780 18611 11 19	55786 18621 12 19	56728 18621 13 19	57553 18621 14 19	58287 18621 15 19	58931 18621 16 19	59483 18621 17 19	59941 18621 18 19	60301 18621 19 19	60572 18621 20 19
20	64400 30701 0 20	56886 25831 1 20	52233 22711 2 20	45740 18851 3 20	40025 16011 4 20	43671 16861 5 20	47713 17831 6 20	50584 18261 7 20	52299 18451 8 20	53876 18541 9 20	55295 18581 10 20	56580 18601 11 20	57795 18611 12 20	58837 18621 13 20	59780 18621 14 20	60607 18621 15 20	61350 18621 16 20	61992 18621 17 20	62550 18621 18 20	63021 18621 19 20	63401 18621 20 20
21	67540 30701 0 21	59409 25831 1 21	54504 22711 2 21	47125 18851 3 21	41518 16011 4 21	45220 16861 5 21	49327 17831 6 21	51995 18261 7 21	53949 18451 8 21	55535 18541 9 21	56880 18581 10 21	58021 18601 11 21	59055 18611 12 21	59998 18621 13 21	60823 18621 14 21	61558 18621 15 21	62201 18621 16 21	62754 18621 17 21	63212 18621 18 21	63571 18621 19 21	63843 18621 20 21
22	70610 30701 0 22	61992 25831 1 22	56775 22711 2 22	48810 18851 3 22	43034 16011 4 22	46797 16861 5 22	50970 17831 6 22	53667 18261 7 22	55632 18451 8 22	57224 18541 9 22	58552 18581 10 22	59715 18601 11 22	60749 18611 12 22	61692 18621 13 22	62517 18621 14 22	63252 18621 15 22	63896 18621 16 22	64448 18621 17 22	64906 18621 18 22	65285 18621 19 22	65537 18621 20 22
23	73680 30701 0 23	64575 25831 1 23	58862 22711 2 23	50533 18851 3 23	44590 16011 4 23	48412 16861 5 23	52652 17831 6 23	55375 18261 7 23	57354 18451 8 23	58932 18541 9 23	60281 18581 10 23	61446 18601 11 23	62482 18611 12 23	63426 18621 13 23	64266 18621 14 23	64986 18621 15 23	65629 18621 16 23	66191 18621 17 23	66680 18621 18 23	67121 18621 19 23	67471 18621 20 23
24	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24
25	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24
26	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24
27	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24
28	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24
29	75743 30701 0 24	66295 25831 1 24	60184 22711 2 24	51764 18851 3 24	45825 16011 4 24	49500 16861 5 24	53802 17831 6 24	56552 18261 7 24	58543 18451 8 24	60148 18541 9 24	61481 18581 10 24	62646 18601 11 24	63682 18611 12 24	64627 18621 13 24	65452 18621 14 24	66187 18621 15 24	66831 18621 16 24	67383 18621 17 24	67841 18621 18 24	68200 18621 19 24	68472 18621 20 24

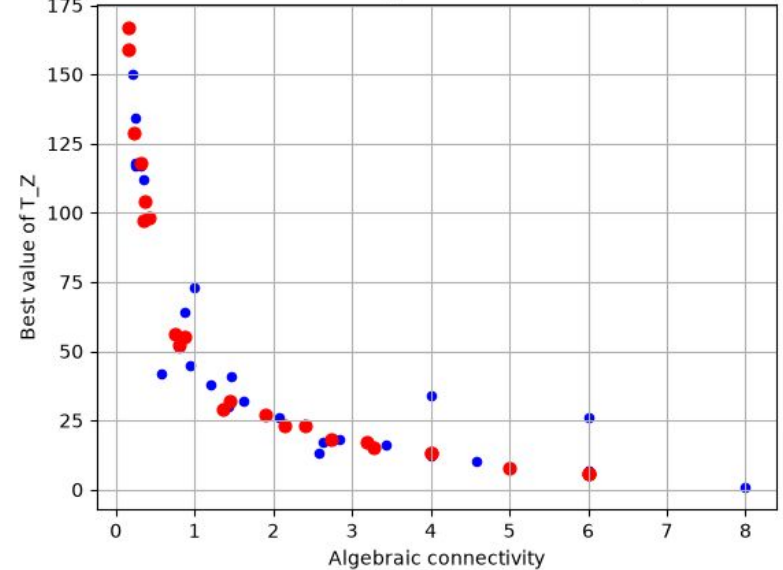
Numerical Experiments - Optimal Iteration Parameters

Algebraic Connectivity

Scatter plot of the Best value of T_Y as a function of Algebraic connectivity



Scatter plot of the Best value of T_Z as a function of Algebraic connectivity



Conclusion and Perspectives

Step 1:

Consensus on vectors $u_{p,m}(t)$ to obtain vector $y_{p,m}(t)$.

Step 2:

Consensus on $D_m D_m^\top y_{p,m}(t)$ to obtain vector $z_{p,m}(t)$.

Estimate quality $\uparrow \Leftrightarrow T_{DAz} \uparrow$.

- Mathematical relation and proof between the algebraic connectivity and the optimal number of iterations of step 2 ?
- Study of the parameters of the update rule ?

Thank You for Listening

References

[1] Lin Xiao and Stephen Boyd. “Fast linear iterations for distributed averaging”. In: Systems & Control Letters 53.1 (2004), pp. 65–78.

[2] Qiuyun Cao et al. “A Novel Iterative Method for Principal Component Analysis”. In: 2026 14th International Electrical Engineering Congress (iEECON). IEEE, 2026. ISBN: 979-8-3315-5925-0.